Fiscal Year:	FY 2010	**Task Last Updated:**	FY 11/15/2010
PI Name:	Edmonds, Jessica Ph.D.		
Project Title:	On Orbit Immuno-Based, Label-Free, White Blood Cell Counting System with MicroElectroMechanical Sensor (MEMS) Technology (OILWBCS-MEMS)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **ISSMP**:ISS Medical Project (archival in 2019)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:			
City:	Cambridge	**State:**	MA
Zip Code:	02142-1189	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	06/28/2010	**End Date:**	06/27/2012
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX10CB07C		
Performance Goal No.:			
Performance Goal Text:			

Aurora Flight Sciences Corporation and partner, Draper Laboratory, propose to develop an on-orbit immuno-based label-free white blood cell counting system using MEMS technology (OILWBCS-MEMS) for human spaceflight experimental and medical monitoring practices. Our proposed system is designed to meet NASA's requirements for a microgravity compatible, miniaturized, light weight peripheral blood cell counting instrument capable of on-orbit cell counting, without high energy lasers, requiring minimal sample volume or exogenous (sheath) fluid, and generating minimal bio-hazardous waste: SBIR topic X14.02 "On Orbit Cell Counting and Analysis Capability". The proposed detection technology leverages changes in optical transmission through a surface due to molecular binding (e.g., antibody-antigen binding). Antibodies specific to the white blood cell surface protein markers (antigens) are pre-coated on the sensor surface to recognize specific white blood cell types with inherently high specificity and sensitivity. In Phase I we developed surface chemistry and demonstrated surface chemistry sensitivity and specificity for total white

Task Description:	blood cells and two lymphocyte subtypes (B-cells, CD4+ T-cells). During phase II we will develop a functional prototype of the OILWBCS-MEMS device to demonstrate that end-to-end operations from sample-in to signal-out produces clinically relevant results. The OILWBCS-MEMS design will include single-use replaceable cartridges for fluid loop and sensor components. POTENTIAL NASA COMMERCIAL APPLICATIONS: The proposed OILWBCS-MEMS white blood cell counting system could provide on orbit white blood cell counting capability, currently not available on orbit, for experimental support and monitoring astronaut health and performing biomedical research on International Space Station (ISS), Crew Exploration Vehicles (CEV), and etc. White blood cell counts are extremely important as a diagnostic tool. With its high specificity/sensitivity, compact size, multiplexed counting capability, and automated operation, the proposed system can also provide a powerful cell counting capability for NASA's ground based biomedical research and development Physiological testing and experimentation will be supported through the use of a white blood cell counting system. White blood cell counts, however, are also extremely important as a diagnostic tool.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Besides NASA's on orbit application, the proposed white blood cell counting system could be used to perform immuno-based cell counting for ground applications, similar to that done by flow cytometers. In general, any existing immuno-based cell counting performed for research and development, could potentially be accomplished with our proposed approach. The benefits of our proposed nanohole array cell counting system include high specificity and sensitivity for simultaneous counting of multiple cell types, compact size, low power consumption, and full automation. The OILWBCS-MEMS will be particularly suited as a tool for remote applications for example in battlefield medical facilities or in other mobile medical centers, such as ambulances, triage centers, disaster response and for use in third-world medical facilities. In these applications, rapid accurate diagnosis is highly desirable, as are ruggedness, portability and containment, all of which are also required features of our OILWBCS-MEMS device.
Task Progress:	New project for FY2010. Reporting not required for this SBIR Phase 2 project. (Note: task added to Task Book in November 2010 when received information.)
Bibliography Type:	Description: (Last Updated:)